

THEORY OF ENTROPICITY (TOE) — LIVING REVIEW LETTERS
SERIES — LETTER IE

PREAMBLE

“Why Information Geometry Is Physical in the Theory of Entropicity (ToE)”

“The Clearest, Deepest Explanation of Why the Fisher-Rao, Fubini-Study, and Amari-Cencov Structures Can Legitimately Be Claimed to Be ‘Physical’ in the Theory of Entropicity (ToE), Even Though They Look Like ‘Statistical’ or ‘Quantum-Information’ Objects at First Glance”

“This is the conceptual bridge forged by the Theory of Entropicity (ToE) between information geometry and the geometry of physical spacetime.”

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The present Preamble is inserted between the Abstract of Letter IE and the main body of the letter. Its purpose is to provide the reader with a self-contained, monograph-grade explanation of why the three principal structures of information geometry — the Fisher-Rao metric, the Fubini-Study metric, and the Amari-Cencov α -connections — are not merely convenient mathematical tools borrowed from statistics and quantum information theory, but are instead the authentic geometric substrates from which the physical universe emerges in the Theory of Entropicity (ToE). This

Preamble is conceptual and philosophical in character rather than derivational; the rigorous mathematical proofs, action principles, and field equations appear in the body of Letter IE and its supplementary appendices. What is offered here is the *why* — the deep justification for the ontological claims that the Theory of Entropicity makes about the physical status of information-geometric structures.

To understand why this Preamble is necessary, one must appreciate the magnitude of the claim. For more than a century, the Fisher information metric has been regarded as a tool of mathematical statistics, the Fubini-Study metric as a tool of quantum information theory, and the α -connections as abstract constructs of differential geometry on statistical manifolds. To declare that these structures are *physical* — that they are the very fabric from which spacetime, matter, and gauge fields emerge — is a claim of extraordinary scope. It demands a correspondingly deep justification. That justification is the purpose of what follows.

The argument proceeds in eight numbered sections, followed by a comprehensive Scholium demonstrating that the assignments of information-geometric structures to physical sectors are not arbitrary choices but are forced by the mathematical nature of each structure. The reader is encouraged to proceed sequentially, as each section builds upon the preceding ones to construct a cumulative case of considerable logical force.

SECTION 1

1. Why Einstein Could Say “Riemannian Geometry Is Spacetime”

1.1 The Radical Identification

Einstein's move was radical because he said:

Riemannian curvature = gravity

Geodesics = inertial motion

Metric = physical distances and times

He took a mathematical structure and declared it ontologically real.

But he could do this because:

Riemannian geometry encodes how distances change.

And physics is fundamentally about how distances and durations change.

Thus, the match was natural.

1.2 The Historical Path: From the Equivalence Principle to the Field Equations

The identification of geometry with gravity did not arrive as a sudden flash of insight. It was the culmination of an eight-year intellectual struggle that ranks among the most extraordinary episodes in the history of science. To appreciate the magnitude of what Einstein achieved — and thereby to understand the precedent upon which the Theory of Entropicity builds — it is essential to trace the key stages of this struggle.

The story begins in 1907, when Einstein, then a patent clerk in Bern, was writing a review article on special relativity. In the course of this work, he contemplated the empirical equality of inertial mass and gravitational mass — a coincidence that Newtonian mechanics accepted without explanation.

Einstein recognized that if a person in a closed box could not distinguish between gravitational acceleration and the acceleration of the box itself, then gravity and acceleration were, in some deep sense, the same thing. This was the *equivalence principle*, and Einstein later called it “the happiest thought of my life.” It was the seed from which general relativity would grow, though the tree would take eight more years to reach maturity.

The equivalence principle immediately implied that gravity could not be described within the framework of special relativity, because special relativity presupposed the existence of global inertial frames — frames that extend uniformly across all of spacetime. But if gravity is equivalent to acceleration, and acceleration is a local phenomenon that varies from point to point, then there can be no global inertial frames in the presence of gravity. The geometry of spacetime itself must vary from point to point. Einstein grasped this implication early, but he did not yet possess the mathematical language to express it.

The critical mathematical breakthrough came in the summer of 1912, when Einstein returned to Zurich and reconnected with his old friend and classmate Marcel Grossmann, who was now a professor of mathematics at the ETH. Einstein told Grossmann that he needed a mathematical framework capable of describing spaces whose geometry varied from point to point. Grossmann recognized immediately that what Einstein needed was the Riemannian geometry that had been developed by Bernhard Riemann in 1854 and subsequently elaborated by Elwin Bruno Christoffel, Gregorio Ricci-Curbastro, and Tullio Levi-Civita into the “absolute differential calculus” — what we now call tensor calculus. The collaboration between Einstein and Grossmann, documented in the famous “Zurich Notebook” of 1912–1913, produced the “Entwurf” (draft) theory, which contained “virtually all essential elements of Einstein’s definitive gravitation theory,” yet fell short of

full general covariance. Einstein spent the next three years wrestling with the problem of covariance, taking what has been described as a “big U-turn” before arriving, in November 1915, at the final field equations of general relativity.

The November 1915 field equations, presented in a series of four communications to the Prussian Academy of Sciences, completed the identification of geometry with gravity. The metric tensor $g_{\mu\nu}$, a purely mathematical object describing the distance structure of a Riemannian manifold, was declared to be the gravitational field itself. The Riemann curvature tensor $R^\rho_{\sigma\mu\nu}$, a purely mathematical object describing how parallel transport around closed loops rotates vectors, was declared to be the measure of gravitational tidal forces. The geodesic equation, a purely mathematical statement about the shortest paths on a curved manifold, was declared to be the equation of motion for freely falling particles. In Einstein’s own unforgettable words: “In the light of knowledge attained, the happy achievement seems almost a matter of course, and any intelligent student can grasp it without too much trouble. But the years of anxious searching in the dark, with their intense longing, their alternation of confidence and exhaustion and the final emergence into light — only those who have experienced it can understand it.”

1.3 The Philosophical Precedent: Riemann’s Habilitationsschrift of 1854

Einstein’s identification of geometry with physics did not emerge from a philosophical vacuum. It had a profound precedent in the work of Bernhard Riemann, whose 1854 *Habilitationsschrift*, “Ueber die Hypothesen, welche der Geometrie zu Grunde liegen” (“On the Hypotheses Which Lie at the Foundations of Geometry”), is one of the most visionary documents in the

history of mathematics. Riemann's lecture, delivered before the faculty of the University of Göttingen on June 10, 1854 — with the aged Carl Friedrich Gauss in attendance, who is reported to have been deeply impressed — accomplished something that no previous mathematician had attempted: it freed geometry from the assumption that the structure of space is fixed and known in advance.

Riemann began by observing that “geometry assumes, as things given, both the notion of space and the first principles of constructions in space.” He then noted that “the propositions of geometry cannot be derived from general notions of magnitude, but that the properties which distinguish space from other conceivable triply extended magnitudes are only to be deduced from experience.” This was a revolutionary statement. It meant that the geometry of physical space is not a logical necessity but an empirical fact — something to be determined by observation and measurement, not by pure reason. Riemann was, in effect, declaring that the metric structure of space is a physical quantity, subject to the same empirical determination as any other physical quantity.

Even more remarkably, Riemann anticipated that the geometry of space might be determined by physical forces. In the concluding section of his lecture, he wrote that “the ground of its metric relations” must be sought “in binding forces which act upon it.” This is, in embryonic form, the central idea of general relativity: the geometry of spacetime is determined by the distribution of matter and energy. Riemann saw this sixty years before Einstein, though he lacked the physical context (the equivalence principle) to make it precise.

The significance of Riemann's insight for the Theory of Entropicity cannot be overstated. Riemann showed that the geometry of space is not fixed *a priori*

but is determined by physical content. Einstein showed that the specific physical content that determines geometry is the stress-energy tensor $T_{\mu\nu}$, representing matter and energy. The Theory of Entropicity takes the next step: it shows that the stress-energy tensor itself, and the spacetime metric it determines, both emerge from a more fundamental information-geometric structure — the Fisher-Rao metric on the statistical manifold of the entropic field. The intellectual trajectory is clear: Riemann $\rightarrow$ Einstein $\rightarrow$ Obidi. Each step deepens our understanding of what geometry is and where it comes from.

1.4 The Kantian Background and Its Demolition

To appreciate the full radicalism of Einstein's move, one must understand what it overthrew. For more than a century before Einstein, the dominant philosophical view of geometry was that of Immanuel Kant, who held in his *Critique of Pure Reason* (1781) that Euclidean geometry was a "synthetic a priori" truth about space. By this, Kant meant that Euclidean geometry was not merely a logical tautology (analytic) but contained genuine information about the world (synthetic), yet this information was known independently of experience (a priori). Space, for Kant, was a "form of intuition" — a necessary framework that the mind imposes on sensory experience. The geometry of this framework was necessarily Euclidean, and no empirical discovery could ever change this.

The development of non-Euclidean geometries by Lobachevsky, Bolyai, and Gauss in the early nineteenth century already challenged Kant's position by showing that Euclidean geometry was not the only logically consistent geometry. But these non-Euclidean geometries were regarded as mathematical curiosities, not as descriptions of physical space. It was Einstein who delivered the decisive blow to the Kantian position by showing

that the geometry of physical spacetime is not Euclidean, not fixed, and not known a priori. The geometry of spacetime is Riemannian (or, more precisely, pseudo-Riemannian with Lorentzian signature), it varies from point to point in response to the distribution of matter and energy, and it can only be determined empirically. Geometry is not a form of intuition imposed by the mind; it is a physical field determined by the content of the universe.

This demolition of the Kantian position is directly relevant to the Theory of Entropicity. The standard view of information geometry — the view that the Fisher-Rao metric is “merely” a statistical tool, that the Fubini-Study metric is “merely” a quantum-information-theoretic construct — plays the same role today that the Kantian view of Euclidean geometry played before Einstein. It assumes that these structures belong to a fixed, abstract, mathematical domain (statistics, information theory) and cannot be physical. The Theory of Entropicity shatters this assumption, just as Einstein shattered Kant’s. The information-geometric structures are not abstract descriptions of our knowledge about physical systems; they are the physical systems themselves, viewed at their most fundamental level.

1.5 Minkowski’s Contribution and the Ontological Promotion of Mathematics

The process by which a mathematical structure becomes physical deserves careful philosophical attention, because it is precisely this process that the Theory of Entropicity replicates for information geometry. A crucial intermediate step was taken by Hermann Minkowski in 1908, who reformulated Einstein’s special relativity in the language of four-dimensional spacetime geometry. In his famous lecture at the 80th Assembly of German Natural Scientists and Physicians in Cologne, Minkowski declared: “Henceforth space by itself, and time by itself, are doomed to fade away into

mere shadows, and only a kind of union of the two will preserve an independent reality.”

Minkowski’s contribution was not merely technical; it was ontological. He took the mathematical structure of a four-dimensional pseudo-Euclidean manifold and declared it to be the arena of physical reality. Space and time, which had been regarded as separate entities since the dawn of human thought, were demoted to “mere shadows” — projections of a deeper four-dimensional reality. This was an ontological promotion of a mathematical structure (four-dimensional geometry) to the status of physical reality, combined with an ontological demotion of previously fundamental entities (separate space and time) to the status of appearances.

The parallel to the Theory of Entropicity is exact. In ToE, spacetime itself — the four-dimensional pseudo-Riemannian manifold of general relativity — is demoted to an emergent phenomenon, a “shadow” of a deeper information-geometric reality. The Fisher-Rao metric on the statistical manifold of the entropic field is promoted to the status of fundamental physical reality, from which the spacetime metric emerges via the emergence map. Just as Minkowski showed that space and time are projections of spacetime, Obidi shows that spacetime is a projection of information geometry. This is the next step in the same intellectual trajectory: the progressive deepening of our understanding of what physical geometry is and where it comes from.

1.6 The Precedent for ToE’s Declaration

The historical analysis of this section establishes a crucial precedent. Before Einstein, geometry was mathematics. After Einstein, geometry was physics. The transition required no new experimental data (the precession of Mercury’s perihelion was already known); what it required was a new *ontological declaration* — the declaration that a mathematical structure

(Riemannian geometry) was identical to a physical structure (the gravitational field). This declaration was validated by its consequences: the correct prediction of light bending, gravitational redshift, gravitational waves, black holes, and the expansion of the universe.

The Theory of Entropicity makes a declaration of the same kind: information geometry is not a description of physics but is the substrate of physics. This declaration, like Einstein's, must be validated by its consequences, and the body of Letter IE demonstrates that it is — through the recovery of general relativity, the derivation of the Einstein field equations, the prediction of the cosmological constant, and the unification of gravity, matter, and electromagnetism from a single entropic field. The present Preamble is concerned not with the derivations themselves but with the *conceptual justification* for the declaration that makes them possible.

SECTION 2

2. What Fisher-Rao and Fubini-Study Actually Measure

2.1 The Key Insight

This is the key insight of the Theory of Entropicity (ToE):

Declaration I: The Fisher-Rao Metric

Fisher-Rao measures distinguishability of probability distributions.

It tells you how “far apart” two states of information are.

In ToE, this metric becomes the amplitude-geometry that gives rise to the emergent spacetime metric.

In ToE, the emergence map $g_s = \lambda g_I$ identifies the Fisher-Rao metric as the pre-geometric structure from which [physical] spacetime geometry emerges.

Declaration II: The Fubini-Study Metric

Fubini-Study measures distinguishability of quantum states.

It tells you how “far apart” two wavefunctions are. [Equivalent to Einstein’s metric]

In ToE, it becomes the dynamical-geometry sector responsible for matter and energy.

In ToE, it encodes the dynamical structure of the entropic field, which manifests as matter content in the emergent spacetime.

Declaration III: The Amari-Cencov α -Connections

Amari-Cencov alpha-connections measure dualistic structure of information flows.

They tell you how “curved” the space of statistical transformations is. [Equivalent to Einstein’s Riemannian curvature]

In ToE, they generate the phase-geometry that becomes gauge structure (including electromagnetism) in the emergent spacetime.

In summary:

Summary

These are not arbitrary metrics.

They measure how information changes.

And physics is fundamentally about how states change.

2.2 The Fisher-Rao Metric: Origin, Definition, and Operational Meaning

The Fisher-Rao metric has a distinguished history. Its roots lie in the work of Ronald Aylmer Fisher, who in 1925 introduced the concept of *Fisher information* in the context of statistical estimation theory. Fisher showed that for a parametric family of probability distributions $\{p(x;\theta)\}$, the precision with which the parameter θ can be estimated from observed data is fundamentally limited by a quantity he called the “information” — what we now denote $I(\theta)$. For a single real parameter, the Fisher information is defined as:

$$I(\theta) = \int p(x;\theta) [\partial \ln p(x;\theta) / \partial \theta]^2 dx$$

The significance of this quantity lies in the Cramér-Rao bound, which states that the variance of any unbiased estimator of θ is bounded below by $1/I(\theta)$. The Fisher information thus quantifies the maximum amount of information that an observable random variable carries about the unknown parameter θ . The greater the Fisher information, the more precisely θ can be estimated;

the smaller the Fisher information, the more “blurred” or indistinguishable nearby values of θ become.

In 1945, the Indian mathematician and statistician Calyampudi Radhakrishna Rao took a decisive step by recognizing that the Fisher information could be generalized to a Riemannian metric on the space of probability distributions. For a family of distributions parametrized by a vector $\theta = (\theta^1, \dots, \theta^n)$, the Fisher-Rao metric is defined as:

$$g_{ab}^{FR}(\theta) = \int p(x;\theta) [\partial \ln p(x;\theta) / \partial \theta^a] [\partial \ln p(x;\theta) / \partial \theta^b] dx$$

This is a genuine Riemannian metric — symmetric, positive-definite, and smoothly varying — on the statistical manifold of probability distributions. It defines a notion of distance on this manifold: the distance between two nearby distributions $p(x;\theta)$ and $p(x;\theta+d\theta)$ is given by $ds^2 = g_{ab}^{FR} d\theta^a d\theta^b$.

The crucial point for the Theory of Entropicity is the *operational meaning* of this distance. The Fisher-Rao distance between two probability distributions is not an arbitrary mathematical construct; it has a precise operational interpretation. It is, up to a constant factor, the minimum number of independent statistical observations required to reliably distinguish between the two distributions. Two distributions that are close in the Fisher-Rao metric are difficult to distinguish; two distributions that are far apart are easy to distinguish. The Fisher-Rao metric thus measures *distinguishability* — the fundamental capacity of one informational state to be differentiated from another.

This operational meaning is what makes the Fisher-Rao metric a candidate for physical significance. In general relativity, the spacetime metric $g_{\mu\nu}$ also measures distinguishability — the distinguishability of spacetime events. The spacetime interval $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ tells you how “far apart” two events are in

terms of physical measurements (proper time for timelike separations, proper distance for spacelike separations). The Theory of Entropicity identifies these two notions of distinguishability: the statistical distinguishability measured by the Fisher-Rao metric *is* the physical distinguishability measured by the spacetime metric, with the emergence map $g_s = \lambda g_I$ providing the quantitative relationship between them.

2.3 The Fubini-Study Metric: Quantum Distinguishability and Dynamical Content

The Fubini-Study metric, named after Guido Fubini (1903) and Eduard Study (1905), is the natural Riemannian metric on complex projective space CP^n . In quantum mechanics, the space of pure states of an $(n+1)$ -dimensional quantum system is precisely CP^n , because quantum states are represented by rays in Hilbert space (vectors differing only by a complex phase represent the same physical state). The Fubini-Study metric is defined, for two infinitesimally separated states $|\psi\rangle$ and $|\psi+d\psi\rangle$, as:

$$ds_{FS}^2 = \langle d\psi/d\psi \rangle - |\langle \psi/d\psi \rangle|^2$$

The second term subtracts the component of $|d\psi\rangle$ that is parallel to $|\psi\rangle$ (which represents a mere phase change and is physically irrelevant), leaving only the component perpendicular to $|\psi\rangle$ (which represents a genuine change in the physical state). The Fubini-Study distance thus measures the “amount of physical change” between two quantum states, with the phase redundancy removed.

Like the Fisher-Rao metric, the Fubini-Study metric has a precise operational interpretation. The Fubini-Study distance between two quantum states $|\psi\rangle$ and $|\phi\rangle$ is related to the probability of distinguishing them by a single optimal quantum measurement: $P_{\text{distinguish}} = \sin^2(d_{FS}(|\psi\rangle, |\phi\rangle))$. States that are close in

the Fubini-Study metric are nearly indistinguishable by any quantum measurement; states that are far apart are easily distinguishable. The Fubini-Study metric is the *quantum* analogue of the Fisher-Rao metric, measuring *quantum distinguishability* rather than classical statistical distinguishability.

In the Theory of Entropicity, the Fubini-Study metric plays a fundamentally different physical role from the Fisher-Rao metric. While the Fisher-Rao metric becomes the spacetime metric (the “container” of physics), the Fubini-Study metric becomes the matter content (the “stuff” of physics). This assignment is not arbitrary but is forced by the mathematical nature of the two metrics. The Fisher-Rao metric is defined on the manifold of probability distributions — it measures how the *amplitude* of the entropic field changes from point to point. The Fubini-Study metric is defined on the space of quantum states — it measures the *dynamical* changes of the complexified entropic field $E = \rho \exp(i\Theta)$. In the Obidi Action, the Fubini-Study sector contributes the kinetic and potential energy terms that, upon emergence, become the stress-energy tensor $T_{\mu\nu}$ in the Einstein field equations.

The connection between the Fubini-Study metric and the Berry phase deserves special mention. The Berry connection $A = i\langle\psi|d\psi\rangle$ and the Berry curvature $F = dA$ are intimately related to the Fubini-Study metric. The real part of the quantum geometric tensor $Q_{ab} = \langle\partial_a\psi|\partial_b\psi\rangle - \langle\partial_a\psi|\psi\rangle\langle\psi|\partial_b\psi\rangle$ gives the Fubini-Study metric, while the imaginary part gives the Berry curvature. In the Theory of Entropicity, this relationship ensures that the dynamical geometry (matter) and the phase geometry (gauge fields) are geometrically unified within the same quantum geometric tensor, while playing distinct physical roles in the emergent spacetime.

2.4 The Amari-Cencov α -Connections: Information Flow and Dual Geometry

The α -connections, introduced systematically by Shun-ichi Amari in his landmark 1985 monograph *Differential-Geometrical Methods in Statistics* and rooted in earlier work by Cencov, constitute a one-parameter family of affine connections on a statistical manifold. Given the Fisher-Rao metric g and the cubic tensor (skewness tensor) $T_{abc} = \int p (\partial_a \ln p)(\partial_b \ln p)(\partial_c \ln p) dx$, the α -connection is defined as:

$$\Gamma^{(\alpha)}_{abc} = \Gamma^{(0)}_{abc} + (\alpha/2) T_{abc}$$

where $\Gamma^{(0)}$ is the Levi-Civita connection of the Fisher-Rao metric. Three values of α are of particular importance: $\alpha = 0$ gives the Levi-Civita connection (the unique torsion-free, metric-compatible connection); $\alpha = +1$ gives the *exponential connection* (e-connection); and $\alpha = -1$ gives the *mixture connection* (m-connection).

The fundamental property of the α -connections is *duality*: the e-connection ($\alpha = +1$) and the m-connection ($\alpha = -1$) are dual with respect to the Fisher-Rao metric, meaning that for any three vector fields X, Y, Z :

$$X \cdot g(Y, Z) = g(\nabla^{(+1)}_X Y, Z) + g(Y, \nabla^{(-1)}_X Z)$$

This duality has deep statistical meaning. The e-connection governs the geometry of *exponential families* — families of distributions that are closed under exponential tilting (such as the Gaussian, Poisson, and binomial families). The m-connection governs the geometry of *mixture families* — families that are closed under convex combination. These two types of families are dual to each other, and the dual connections capture this duality geometrically.

In the Theory of Entropicity, the α -connections play the role of gauge structure. The phase $\Theta(x)$ of the complexified entropic field $E = \rho \exp(i\Theta)$ transforms under phase rotations $\Theta \rightarrow \Theta + \chi$ as a U(1) connection. The dual structure of the α -connections — the e-connection governing the coupling of the phase to matter (source terms) and the m-connection governing the dual magnetic formulation — becomes the gauge structure of electromagnetism in the emergent spacetime. The α -connections are not metric connections (they are not compatible with the Fisher-Rao metric for $\alpha \neq 0$), which is precisely why they cannot play the role of the spacetime connection (which must be metric-compatible). Instead, they encode the additional geometric structure — the gauge structure — that goes beyond the spacetime metric.

The fact that the α -connections are *affine* (connection-like) rather than *metric* (distance-like) is the mathematical reason they become gauge connections rather than spacetime structure. In physics, gauge connections are also affine connections — they tell you how to parallel-transport charged fields along curves, but they do not define distances. The α -connections do exactly the same thing on the statistical manifold: they tell you how to parallel-transport probability distributions along curves in parameter space, but they do not define distances (that is the job of the Fisher-Rao metric). The correspondence between information-geometric structure and physical structure is thus precise and detailed, not vague or metaphorical.

SECTION 3

3. The Leap of the Theory of Entropicity

3.1 The Declaration

Information geometry is not a description of physics.
It is the substrate of physics.
This is the same kind of leap Einstein made.

Einstein:

“Geometry is physical.”

— versus —

ToE:

“Information geometry is physical.”

Why did the Theory of Entropicity (ToE) have to make this declarative leap?

The Ontological Foundation

Because:

Every physical system is fundamentally a system of information.

Every physical evolution is fundamentally a change in information.

Every physical interaction is fundamentally an exchange of information.

Hence, the Theory of Entropicity (ToE) declares that the geometry that measures information change is the geometry that determines physical change.

This is the core and, at the same time, the elegance of Obidi’s Theory of Entropicity (ToE).

3.2 Ontological Promotion: The Philosophical Concept

The philosophical operation at the heart of the Theory of Entropicity is what may be called *ontological promotion*: taking a mathematical structure that was previously regarded as a tool for *describing* reality and declaring it to *be* reality. This operation is not new in the history of physics — Einstein performed it when he promoted Riemannian geometry from a mathematical framework to the physical substance of gravity — but each instance of it is radical and requires deep justification.

The justification for ontological promotion cannot be purely mathematical. Mathematics alone cannot tell you that a particular mathematical structure is physically real; that is a claim about the world, not about mathematics. The justification must come from the *match* between what the mathematical structure measures and what physics requires. Einstein's justification was that Riemannian geometry measures distances and curvature, and physics is about distances, durations, and the forces (gravity) that affect them. The Theory of Entropicity's justification is that information geometry measures distinguishability, change, and flow, and physics is fundamentally about distinguishable states, their changes, and their interactions.

The difference between the standard view and the ToE view can be stated sharply. The *standard view* treats information as *epistemological* — information is what we know about a physical system. The physical system exists independently of our knowledge of it; information is a feature of our description, not of the system itself. In this view, the Fisher-Rao metric measures the precision of our knowledge about a system, and the Fubini-Study metric measures the distinguishability of our quantum descriptions. These are properties of our epistemic relation to the system, not properties of the system itself.

The *ToE view* treats information as *ontological* — information is what the physical system *is*. There is no physical system “behind” or “beneath” the information; the information is the substance of the system. In this view, the Fisher-Rao metric measures real, physical distances (the distances between different configurations of the entropic field), and the Fubini-Study metric measures real, physical dynamical content (the kinetic and potential energy of the entropic field). The transition from the epistemological view to the ontological view is the central conceptual move of the Theory of Entropicity.

3.3 Wheeler’s “It from Bit” and Landauer’s Principle

The Theory of Entropicity realizes concretely what John Archibald Wheeler envisioned programmatically in his “It from Bit” doctrine. Wheeler proposed in 1989 that “every *it* — every particle, every field of force, even the spacetime continuum itself — derives its function, its meaning, its very existence entirely — even if in some contexts indirectly — from the apparatus-elicited answers to yes-or-no questions, binary choices, *bits*.” Wheeler’s vision was bold but programmatic: he did not provide the mathematical framework to realize it. The Theory of Entropicity provides that framework. The “bits” of Wheeler’s vision are the probability distributions on the statistical manifold; the “its” are the spacetime events, matter configurations, and gauge fields that emerge from the entropic field via the emergence map and the Obidi Action.

A closely related precedent is Rolf Landauer’s principle, first stated in 1961, which establishes that information processing has a minimum thermodynamic cost: the erasure of one bit of information dissipates at least $k_B T \ln 2$ of energy as heat. Landauer’s principle demonstrates that information is *physical* in the minimal sense that information processing is constrained by the laws of thermodynamics. This principle was

experimentally verified in 2012 by Bérut et al., who measured the heat dissipated by the erasure of a single bit in a colloidal particle system and confirmed that it saturates at the Landauer bound in the limit of slow erasure.

The Theory of Entropicity goes far beyond Landauer's principle. Landauer showed that information is physical in the sense that it has thermodynamic consequences. ToE shows that information is physical in the far stronger sense that it is the *foundation* of physics — that spacetime, matter, and gauge fields are all manifestations of information-geometric structures. Landauer's principle is a corollary of this deeper truth: information has thermodynamic consequences *because* information is the substance from which thermodynamics (and all of physics) emerges.

3.4 The Logical Structure: A Syllogism, Not an Analogy

It is essential to understand that the Theory of Entropicity's claim is not an analogy. It is not saying that information geometry is “like” physical geometry, or that information-geometric structures “behave as if” they were physical. It is making a literal identification, grounded in a logical syllogism:

Premise 1: Every physical system is fundamentally a system of information. (This is supported by quantum mechanics, where the state of a system is completely specified by its wavefunction — an informational object — and by the holographic principle, which bounds the information content of a region by its boundary area.)

Premise 2: The natural geometry of information is the Fisher-Rao/Fubini-Study/ α -connection geometry. (This is a mathematical fact: the Fisher-Rao metric is the unique Riemannian metric on statistical manifolds invariant under sufficient statistics, and the Fubini-Study metric is the unique Kähler metric on the quantum state space.)

Conclusion: The geometry of physics IS the geometry of information.

This is a deductive argument, not an inductive one. If both premises are true, the conclusion follows necessarily. The Theory of Entropicity asserts both premises and draws the conclusion. The remainder of this Preamble, and the body of Letter IE, provide the evidence and the mathematical framework that justify both premises and demonstrate the consequences of the conclusion.

SECTION 4

4. Why Fisher-Rao Becomes Spacetime in ToE

4.1 The Identification

The Fisher-Rao metric measures:
How distinguishable two nearby entropic configurations are.

In the Theory of Entropicity (ToE), we make the following declarations based on the above Fisher-Rao metric measures:

The Fisher-Rao Declarations

“Nearby entropic configurations” = “nearby physical states”

“Distinguishability” = “physical separation”

“Curvature of information” = “curvature of spacetime”

Consequences

Thus:

Spacetime distance = entropic distinguishability.

Spacetime curvature = entropic curvature.

This is not metaphor.

This is the emergence map (EM):

$$\mathbf{g}_s = \lambda \mathbf{g}_I$$

This is the Theory of Entropicity (ToE) analogue of Einstein's identification of gravity with curvature.

4.2 The Emergence Map: Physical Meaning

The emergence map $\mathbf{g}_s = \lambda \mathbf{g}_I$ is the most fundamental equation in the Theory of Entropicity. It states that the spacetime metric $\mathbf{g}_s$ (the object that Einstein identified with the gravitational field) is proportional to the Fisher-Rao information metric $\mathbf{g}_I$ (the object that measures statistical distinguishability on the entropic manifold). The constant of proportionality λ converts between information-geometric units and spacetime units.

What does this equation mean physically? It means that the distance between two spacetime events — the quantity that determines how clocks tick, how rulers measure, how light propagates, how gravity acts — is nothing more, and nothing less, than the statistical distinguishability of the corresponding entropic configurations, scaled by the universal conversion factor λ . Two

events that are far apart in spacetime correspond to entropic configurations that are highly distinguishable; two events that are close together correspond to configurations that are nearly indistinguishable. The entire fabric of spacetime — its distances, its curvature, its causal structure — is a macroscopic image of the information-geometric structure of the entropic field.

The emergence map is formally a pullback of the spacetime metric along the emergence morphism $\Phi: M_I \rightarrow M_S$, written more precisely as $\Phi^*g_S = \lambda g_I$. This means that the emergence morphism is a conformal immersion (up to the constant λ) of the information manifold into the spacetime manifold. The information manifold is the “microscopic” structure; the spacetime manifold is the “macroscopic” image. The emergence map tells you how to reconstruct the macroscopic image from the microscopic structure.

The connection to the Curvature Transfer Theorem of Letter IE is direct and profound. The Curvature Transfer Theorem states that $R^S = \lambda \Phi^*(R^I) + O(1/N)$, where R^S is the spacetime Riemann curvature, R^I is the information-geometric curvature, and N is the number of entropic degrees of freedom. This theorem shows that spacetime curvature is a direct, quantitative image of information-geometric curvature, with corrections that vanish in the thermodynamic limit ($N \rightarrow \infty$). General relativity — the theory that spacetime curvature equals gravitational field — is thus a macroscopic consequence of the more fundamental identification of information-geometric curvature with the entropic structure of reality.

4.3 The Cencov Uniqueness Theorem

One of the most powerful arguments for the physical significance of the Fisher-Rao metric in the Theory of Entropicity is the Cencov uniqueness theorem, proved by Nikolai Nikolaevich Cencov (Chentsov) in his 1972

monograph *Statistical Decision Rules and Optimal Inference*. The theorem states:

On the manifold of probability distributions over a finite sample space, the Fisher-Rao metric is the unique Riemannian metric — up to an overall constant scalar factor — that is invariant under Markov morphisms (information-preserving stochastic transformations of the data).

The importance of this theorem for the Theory of Entropicity cannot be overstated. It means that the Fisher-Rao metric is not *one possible choice* among many Riemannian metrics on the statistical manifold; it is the *only choice* that respects the fundamental symmetries of information processing. Any other Riemannian metric would distinguish between different representations of the same information — it would measure not the information itself but the labels we attach to it. Only the Fisher-Rao metric is intrinsic to the information content, independent of how the information is represented.

This removes what might otherwise be a devastating objection to the Theory of Entropicity: the objection that the choice of the Fisher-Rao metric as the pre-geometric structure of spacetime is arbitrary, that one could equally well have chosen some other metric and obtained a different theory. Cencov's theorem shows that this objection fails. There is no other metric to choose. The Fisher-Rao metric is as unique among information-geometric metrics as the Levi-Civita connection is among metric-compatible torsion-free connections, or as the Einstein tensor is among divergence-free second-order symmetric tensors formed from the metric. The Theory of Entropicity inherits the full force of this mathematical uniqueness.

4.4 The Scholium on λ

Scholium on the ToE Metric Conversion Factor of λ

This emergence map ($\Phi^*g_s = \lambda g_i$) is the Theory of Entropicity (ToE) analogue of Einstein's identification of gravity with curvature.

In the Theory of Entropicity (ToE), the conversion factor λ must be a constant because it is the universal conversion factor between the Fisher-Rao information metric and the emergent spacetime metric. That is, λ is a conversion factor between information geometry and spacetime geometry. If λ were a function, the emergence map would violate the equivalence principle, break the uniqueness of the Fisher-Rao metric, introduce unphysical forces, and fail to reproduce general relativity. The constancy of λ is therefore required by both the mathematics of information geometry and the physics of spacetime emergence.

If λ were a function (a variable, and not a constant), the emergence map would break general relativity.

In ToE, free motion corresponds to Fisher-Rao geodesics, where λ must be a constant.

In all of the above, λ must be constant because of Cencov's theorem. Cencov's theorem states: The Fisher-Rao metric is unique up to a constant multiplicative factor. This is a deep result.

In the Theory of Entropicity, λ is fixed at approximately 10^{-47} by the entropic curvature gap between the highly curved Fisher-Rao information manifold and the extremely flat physical spacetime. This

value is not a range and not a repetition of existing literature; it is uniquely determined by the emergence map and required for the recovery of general relativity, the equivalence principle, and the observed cosmological constant.

Although the information manifold curvature R_I and the spacetime curvature R_S each vary over ranges, the emergence constant λ is not defined by pointwise curvature values. Instead, λ is fixed by the ratio of the characteristic curvature scales of the two manifolds. This ratio is sharply determined by the entropic curvature gap and yields λ approximately equals 10^{-47} . If λ were allowed to vary over the ranges of R_I or R_S , the equivalence principle would fail, general relativity would not emerge, and the observed universe could not exist. Thus, λ is a universal conversion constant, not a variable.

4.5 The Value of λ : Dimensional Analysis and Physical Meaning

The emergence constant λ can be computed from fundamental constants. The Fisher-Rao metric has dimensions of inverse information (or, equivalently, inverse entropy), while the spacetime metric has dimensions of length squared. The conversion between the two requires a constant with dimensions of $[\text{length}^2 \times \text{entropy}]$. From the fundamental constants of nature, the unique combination with these dimensions is:

$$\lambda = l_P^2 / (4 k_B) = \hbar G / (4 k_B c^3)$$

where $l_P = (\hbar G/c^3)^{1/2} \approx 1.616 \times 10^{-35}$ m is the Planck length, $\hbar$ is the reduced Planck constant, G is Newton's gravitational constant, c is the speed of light, and k_B is Boltzmann's constant. Substituting numerical values:

$$\lambda \approx (1.616 \times 10^{-35})^2 / (4 \times 1.381 \times 10^{-23}) \approx 10^{-47} \text{ m}^2/\text{J}\cdot\text{K}^{-1}$$

This extraordinarily small value has a profound physical meaning: the information manifold is enormously more curved than physical spacetime. A curvature of order unity on the information manifold (a perfectly natural value in information-geometric terms) corresponds to a curvature of order 10^{-47} in spacetime — an almost perfectly flat spacetime, which is exactly what we observe. The physical universe is, in the language of the Theory of Entropicity, an extremely “zoomed out” and “smoothed out” image of the underlying information manifold. The tiny value of λ is the quantitative expression of this enormous scale separation.

The four consequences of allowing λ to vary deserve detailed exposition:

1. **Violation of the equivalence principle:** If λ varied from point to point, then the relationship between information-geometric distances and spacetime distances would also vary from point to point. This would mean that freely falling particles in the same gravitational field would follow different trajectories depending on their information content — a direct violation of the universality of free fall (the weak equivalence principle). All experimental tests of the equivalence principle, from Eötvös to MICROSCOPE, confirm universality to extraordinary precision.
2. **Breaking the uniqueness of the Fisher-Rao metric:** Cencov's theorem states that the Fisher-Rao metric is unique up to a *constant* multiplicative factor. If λ were a function, the emergent metric $g_s = \lambda(x) g_I$ would not be a constant multiple of the Fisher-Rao metric; it

would be a conformally rescaled version. But a conformally rescaled Fisher-Rao metric is, in general, not invariant under Markov morphisms — it violates the very property that makes the Fisher-Rao metric unique. The emergence map would thus be inconsistent with the mathematical structure it is built upon.

3. **Introduction of unphysical forces:** A variable λ would introduce a scalar field $\lambda(x)$ into the theory, analogous to the Brans-Dicke scalar field. This scalar field would mediate a fifth force, coupling to matter in ways that are not observed. The stringent experimental limits on fifth forces (from torsion-balance experiments, lunar laser ranging, and binary pulsar timing) rule out a variable λ at the levels required for observable effects.
4. **Failure to reproduce general relativity:** The recovery of the Einstein field equations from the Obidi Action requires that the emergence map be a constant-factor identification. If λ varied, the emergent field equations would contain additional terms involving the gradient and Laplacian of λ , yielding a scalar-tensor theory rather than general relativity. Since general relativity is extraordinarily well-tested in the weak-field and strong-field regimes, any deviation from it is strongly constrained.

SECTION 5

5. Why Fubini-Study Becomes Matter in ToE

5.1 The Declarations

The Fubini-Study metric measures:
How dynamical quantum states differ.

In the Theory of Entropicity (ToE), we make the following declarations based on the above Fubini-Study metric measures:

The Fubini-Study Declarations

Matter = dynamical structure of the entropic field

Mass = resistance to change in entropic configuration

Energy = curvature of the Fubini-Study sector

Thus:

Matter is the dynamical geometry of the entropic field.

This is why the Fubini-Study metric appears in the dynamical part of the Obidi Action.

5.2 The Fubini-Study Metric on Projective Space

To understand why the Fubini-Study metric becomes the matter sector, one must first appreciate its mathematical construction in detail. Complex projective space CP^n is the space of lines through the origin in C^{n+1} . If we represent a point in CP^n by homogeneous coordinates $[z^0 : z^1 : \dots : z^n]$, the Fubini-Study metric can be written in the affine chart $z^0 \neq 0$, with inhomogeneous coordinates $w^k = z^k/z^0$, as:

$$ds_{FS}^2 = [(1 + |w|^2)\delta_{jk} - \bar{w}_j w_k] / (1 + |w|^2)^2 \cdot dw^j d\bar{w}^k$$

This metric is Kähler (it derives from a Kähler potential $K = \ln(1 + |w|^2)$), it has constant positive holomorphic sectional curvature, and it is the unique (up to a constant factor) Kähler metric on CP^n that is invariant under the

natural action of the unitary group $U(n+1)$. These properties make it the canonical metric on the space of quantum states.

The Fubini-Study metric is intimately related to the Berry connection and Berry curvature. The quantum geometric tensor, defined for a family of states $|\psi(\theta)\rangle$ parametrized by $\theta = (\theta^1, \dots, \theta^n)$, is $Q_{ab} = \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle$. The real (symmetric) part of Q_{ab} is the Fubini-Study metric, and the imaginary (antisymmetric) part is one-half the Berry curvature. This decomposition shows that the geometry of the quantum state space naturally splits into a metric part (Fubini-Study) and a connection part (Berry), corresponding precisely to the matter sector and the gauge sector in the Theory of Entropicity.

5.3 Why Matter = Dynamical Geometry

In standard physics, matter is described by fields — scalar fields, spinor fields, vector fields — that propagate on a fixed (or dynamical) spacetime background. The fields satisfy equations of motion (Klein-Gordon, Dirac, Maxwell) that determine how they evolve in time and space. The energy and momentum carried by these fields are encoded in the stress-energy tensor $T_{\mu\nu}$, which serves as the source for Einstein's field equations.

In the Theory of Entropicity, this picture is deepened. Matter is not a separate entity placed on top of spacetime; it is the *dynamical structure* of the entropic field itself. The complexified entropic field $E = \rho \exp(i\Theta)$ has two sectors: the amplitude sector ρ , whose geometry (measured by the Fisher-Rao metric) becomes spacetime, and the dynamical fluctuations of the full field E , whose geometry (measured by the Fubini-Study metric) becomes matter. The Fubini-Study metric measures the “speed” at which the entropic field configuration changes as one moves through parameter space. Where the field is changing rapidly (large Fubini-Study distance), there is a large

amount of dynamical activity, which manifests as matter and energy in the emergent spacetime. Where the field is static or slowly varying (small Fubini-Study distance), there is vacuum.

5.4 Why Mass = Resistance to Entropic Change

In Newtonian mechanics, mass is defined as the resistance to acceleration: $F = ma$. An object with large mass requires a large force to change its velocity. In the Theory of Entropicity, this concept acquires a deeper meaning. Mass is the resistance to change in the entropic configuration — the “inertia” of the entropic field. This resistance is determined by the curvature of the Fubini-Study metric in the amplitude direction.

To see why, consider a configuration of the entropic field with a large Fubini-Study curvature in the amplitude sector. This means that the field’s quantum state is “stiff” — small changes in the parameters produce large changes in the state, requiring a large “force” (in the information-geometric sense) to effect the change. In the emergent spacetime, this stiffness manifests as inertial mass. Conversely, a configuration with small Fubini-Study curvature is “flexible” — it can be easily changed, and it corresponds to a particle with small mass or to vacuum.

This identification provides a geometric explanation for the otherwise mysterious property of mass. In the Standard Model, the masses of elementary particles are determined by their coupling to the Higgs field — but the Higgs mechanism does not explain *why* particles have mass; it only provides the mechanism. The Theory of Entropicity offers a deeper answer: particles have mass because the entropic field configurations that constitute them have Fubini-Study curvature. Mass is geometry — specifically, it is the curvature of the Fubini-Study sector of the information manifold.

5.5 Why Energy = Fubini-Study Curvature

Energy is the capacity to do work — that is, the capacity to change the state of a system. In the Theory of Entropicity, the capacity to change the entropic state is measured by the curvature of the Fubini-Study metric. A flat Fubini-Study metric means that the entropic field is in a state of minimum dynamical activity — this is the vacuum, with zero energy density. A curved Fubini-Study metric means that the entropic field has dynamical content — this is matter, with nonzero energy density.

The connection to the entropic stress-energy tensor, developed in the supplementary appendices of Letter IE, makes this precise. The entropic stress-energy tensor $T_{\mu\nu}^{(\text{ent})}$ is constructed from the Fubini-Study sector of the Obidi Action by variation with respect to the emergent spacetime metric. Its components encode the energy density, momentum density, pressure, and stress of the entropic field in the emergent spacetime. The trace of this tensor gives the energy density, which is proportional to the scalar curvature of the Fubini-Study metric evaluated on the entropic field configuration. This establishes the identification: energy = Fubini-Study curvature, not as a metaphor but as a mathematical equality.

SECTION 6

6. Why Amari-Cencov α -Connections Become Electromagnetism

6.1 The Declarations

The α -connections measure:
How information flows twist and turn.

In the Theory of Entropicity (ToE), we make the following declarations based on the above alpha-connections measure:

The α -Connection Declarations

Phase structure = gauge structure

α = plus or minus 1 correspond to dual connections

The U(1) gauge field emerges from the phase geometry

Thus:

Electromagnetism is the phase geometry of the entropic field.

6.2 The Mathematical Structure of the α -Connections

The α -connections form a one-parameter family indexed by the real parameter α . Recall the defining equation: $\Gamma^{(\alpha)}_{abc} = \Gamma^{(0)}_{abc} + (\alpha/2) T_{abc}$, where $\Gamma^{(0)}$ is the Levi-Civita connection of the Fisher-Rao metric and T_{abc} is the Amari-Cencov cubic tensor (also called the skewness tensor). The Levi-Civita connection $\Gamma^{(0)}$ is the unique connection that is both torsion-free and metric-compatible. For $\alpha \neq 0$, the α -connection is no longer metric-compatible: it does not preserve the Fisher-Rao inner product under parallel transport. This failure of metric compatibility is not a defect; it is a feature. It is precisely what allows the α -connections to encode structure that is *additional to* the metric structure — namely, gauge structure.

The duality between the e-connection ($\alpha = +1$) and the m-connection ($\alpha = -1$) is one of the most beautiful features of information geometry. It reflects the fundamental duality between two ways of combining probability distributions: exponential mixing (which corresponds to Bayesian updating,

adding log-likelihoods) and convex mixing (which corresponds to forming ensembles, averaging probabilities). These two operations are dual in a precise sense: if you parametrize the statistical manifold using natural parameters (the parameters in which the exponential family is linear), the e-connection becomes flat; if you parametrize using expectation parameters (the means of sufficient statistics), the m-connection becomes flat. The two flat coordinate systems are dual, and the statistical manifold has the structure of a *dually flat manifold* — a concept that has no analogue in standard Riemannian geometry.

6.3 Why Phase Geometry = Gauge Structure

The phase $\Theta(x)$ of the complexified entropic field $E = \rho \exp(i\Theta)$ plays a role analogous to the phase of a charged quantum field in ordinary quantum field theory. Under a local phase rotation $\Theta(x) \rightarrow \Theta(x) + \chi(x)$, the amplitude ρ is unchanged but the phase shifts. To maintain the invariance of the Obidi Action under such local phase rotations, one must introduce a compensating gauge field A_μ that transforms as $A_\mu \rightarrow A_\mu + \partial_\mu \chi$. This is exactly the mechanism by which gauge fields arise in standard quantum field theory (the Weyl-Yang-Mills construction), but in the Theory of Entropicity, it arises from the phase sector of the entropic field rather than being postulated independently.

The connection to the α -connections is as follows. The e-connection ($\alpha = +1$) governs how probability distributions transform under exponential tilting — that is, under the addition of a linear function to the log-likelihood. In the entropic field, this corresponds to the addition of a phase gradient to the entropic potential, which is precisely a gauge transformation. The m-connection ($\alpha = -1$) governs the dual transformation — the mixture operation — which corresponds to the dual (magnetic) formulation of the gauge field. The duality between the e-connection and the m-connection thus becomes the

electromagnetic duality between the electric and magnetic formulations of Maxwell's equations.

The entropic four-potential, as derived in Letter IIA of the ToE Living Review Letters series, takes the form $A_\mu = (\hbar_{\text{ent}}/q_{\text{ent}}) \partial_\mu \Theta$, where $\hbar_{\text{ent}}$ is the entropic Planck constant and q_{ent} is the entropic charge. This transforms exactly as an electromagnetic gauge potential under phase rotations. The field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the curvature of this connection, and it satisfies Maxwell's equations as a consequence of the equations of motion derived from the Obidi Action. Electromagnetism is thus not an independent force appended to the theory; it emerges naturally from the phase geometry of the entropic field, which is governed by the α -connections of information geometry.

6.4 Why the α -Connections Cannot Be the Spacetime Connection

It is important to understand why the α -connections become gauge connections rather than the spacetime (Levi-Civita) connection. The reason is mathematical and sharp. The Levi-Civita connection is the unique connection that is both torsion-free and metric-compatible. The α -connections (for $\alpha \neq 0$) are torsion-free but *not* metric-compatible. In general relativity, the spacetime connection must be metric-compatible, because metric compatibility ensures that the lengths of vectors are preserved under parallel transport — which is the geometric expression of the conservation of energy and momentum. A non-metric-compatible connection would allow the “length” of a vector to change as it is transported, which would correspond to a violation of energy conservation.

Gauge connections, by contrast, are *not* required to be metric-compatible. A gauge connection (such as the electromagnetic four-potential A_μ) tells you

how to parallel-transport the phase of a charged field, not the length of a spacetime vector. The phase can rotate freely under parallel transport; there is no conservation law that constrains it (the conservation of charge is a separate matter, arising from the gauge symmetry itself, not from metric compatibility). The α -connections, being non-metric-compatible affine connections, thus have exactly the right mathematical structure to serve as gauge connections and exactly the wrong mathematical structure to serve as the spacetime connection.

SECTION 7

7. The Unified Picture

7.1 The Unification

Einstein

unified:

geometry

gravity

The Theory of Entropicity (ToE) Unifies:

amplitude geometry $\rightarrow$ spacetime

dynamical geometry $\rightarrow$ matter

phase geometry $\rightarrow$ electromagnetism

All from one entropic field.

This is why one can legitimately say:

Riemann is to GR what Fisher-Rao, Fubini-Study, and Amari-Cencov are to ToE.

Because:

Riemannian geometry encodes physical distances.

Fisher-Rao encodes entropic distances.

Fubini-Study encodes dynamical distances.

alpha-connections encode phase/gauge distances.

And in ToE:

Physical reality = entropic reality.

Hence, entropic geometry = physical geometry.

7.2 The History of Unification in Physics

The history of physics is, in large measure, a history of unification — the discovery that phenomena previously regarded as separate are, in fact, manifestations of a single underlying structure. Each major unification has deepened our understanding of nature and simplified the conceptual landscape of physics.

The first great unification was achieved by Isaac Newton in 1687, who showed that the force that causes an apple to fall from a tree and the force that holds the Moon in its orbit around the Earth are the same force: gravity. Before Newton, celestial mechanics and terrestrial mechanics were regarded as separate domains governed by different laws. Newton unified them under a single law of universal gravitation.

The second great unification was achieved by James Clerk Maxwell in the 1860s, who showed that electricity and magnetism — previously regarded as distinct phenomena — are aspects of a single entity: the electromagnetic field. Maxwell’s equations unified Coulomb’s law, Ampère’s law, Faraday’s law, and the absence of magnetic monopoles into a single, elegant framework, and predicted the existence of electromagnetic waves (light).

The third great unification was Einstein’s unification of space and time into spacetime (special relativity, 1905) and of spacetime geometry with gravity (general relativity, 1915). The fourth was the electroweak unification of Sheldon Glashow, Abdus Salam, and Steven Weinberg in the 1960s and 1970s, which showed that the electromagnetic force and the weak nuclear force are aspects of a single electroweak force at high energies.

The Theory of Entropicity represents a unification that goes deeper than all of these. Previous unifications united different forces or different aspects of spacetime. ToE unifies the very *categories* of physics — geometry (spacetime), matter (the stuff that inhabits spacetime), and gauge fields (the forces that act on matter) — into a single information-geometric structure. These three categories, which have been regarded as fundamentally distinct since the formulation of general relativity, are revealed by ToE to be three aspects of a single entropic field, distinguished by the type of information-geometric structure they correspond to.

7.3 Comparison Table: General Relativity vs. Theory of Entropicity

Feature	Einstein’s General Relativity	Theory of Entropicity (ToE)
Fundamental structure	Riemannian (pseudo-Riemannian) geometry	Information geometry (Fisher-Rao, Fubini-

Feature	Einstein’s General Relativity	Theory of Entropicity (ToE)
		Study, α -connections)
Number of geometric structures	One (the metric $g_{\mu\nu}$)	Three (metric, dynamical, affine)
Physical sectors	One (gravity/spacetime)	Three (spacetime, matter, gauge fields)
Fundamental field	Metric tensor $g_{\mu\nu}$	Entropic field $E = \rho \exp(i\Theta)$
Matter treatment	External source ($T_{\mu\nu}$ postulated)	Emergent from dynamical geometry
Gauge fields	Not included (must be added separately)	Emergent from phase geometry
Ontological declaration	“Geometry is physical”	“Information geometry is physical”
Unification scope	Gravity and spacetime geometry	Spacetime, matter, and electromagnetism

This comparison makes clear the scope of the unification achieved by the Theory of Entropicity. Where Einstein required one geometric structure to describe one physical sector (gravity), the Theory of Entropicity uses three related geometric structures to describe three physical sectors, all emerging from a single entropic field. The simplicity of the underlying structure — one field, three geometric aspects — combined with the richness of its physical consequences — spacetime, matter, and electromagnetism — is the hallmark of a deep unification.

SECTION 8

8. The Final Clarity

8.1 The Natural Match

The Theory of Entropicity (ToE) doesn't need to "force" Fisher-Rao or Fubini-Study to be physical.

That is because these structures already measure the right thing that physics requires:

What Information Geometry Measures

- change
- distinguishability
- curvature
- flow
- structure

And since Physics is nothing but the evolution of [distinguishable] states, it then necessarily follows that the geometry that measures distinguishability is the geometry of physics.

Einstein replaced forces with geometry.

The Theory of Entropicity (ToE) is replacing that Einstein's geometry with information geometry.

This is the next conceptual step in the formulation of the Theory of Entropicity (ToE).

8.2 The Logical Chain: Forces → Geometry → Information Geometry

The intellectual trajectory of fundamental physics can be understood as a series of *ontological deepenings*, each of which reveals the previously fundamental layer of reality to be emergent from a deeper one:

Stage 1 (Newton, 1687): The world consists of particles moving under forces. Forces are the fundamental explanation for motion. Space and time are fixed, absolute backgrounds in which particles and forces operate.

Stage 2 (Einstein, 1915): Forces (at least gravity) are not fundamental. Gravity is not a force but a manifestation of spacetime curvature. Geometry replaces forces as the fundamental explanation for gravitational motion. But spacetime geometry is now dynamical, not fixed — it is a physical field determined by the distribution of matter and energy.

Stage 3 (Obidi, ToE): Spacetime geometry is not fundamental. Spacetime itself is an emergent phenomenon, arising from the more fundamental information-geometric structure of the entropic field. Information geometry replaces spacetime geometry as the deepest layer of physical reality.

Each stage deepens our understanding of what is fundamental and what is emergent. Newton's forces are emergent from Einstein's geometry. Einstein's geometry is emergent from Obidi's information geometry. At each stage, the previously "fundamental" entity is revealed to be a macroscopic manifestation of a deeper structure.

Crucially, each stage *contains* the previous one as a limiting case. General relativity contains Newtonian gravity as the weak-field, low-velocity limit. The Theory of Entropicity contains general relativity as the classical, macroscopic limit (the limit $N \rightarrow \infty$, where the $O(1/N)$ corrections in the Curvature Transfer

Theorem vanish). The Theory of Entropicity does not *replace* general relativity; it *reveals what general relativity is made of*. Just as knowing that water is made of H₂O molecules does not replace the fluid dynamics of water but deepens our understanding of why fluid dynamics works, knowing that spacetime is made of information-geometric structure does not replace general relativity but deepens our understanding of why general relativity works.

8.3 “The Next Conceptual Step”: What It Means

When the Theory of Entropicity describes itself as “the next conceptual step,” it is not claiming to be a minor modification or extension of existing theory. It is claiming to be the next step in a historical sequence of ontological revolutions, each of which has transformed our understanding of the nature of physical reality. The first revolution (Newton) showed that celestial and terrestrial phenomena obey the same laws. The second (Maxwell) showed that electricity and magnetism are aspects of one field. The third (Einstein) showed that gravity is geometry. The fourth (Obidi) shows that geometry is information geometry — that the fabric of spacetime, the substance of matter, and the structure of gauge fields all emerge from a single entropic field whose geometry is the geometry of information.

The depth of this claim is matched by the scope of its consequences. If the Theory of Entropicity is correct, then every equation in general relativity, every term in the Standard Model Lagrangian, every prediction of quantum field theory is ultimately a statement about information geometry. The laws of physics are not laws imposed upon nature from outside; they are the geometric properties of the information-geometric manifold on which reality is defined. Physics is information geometry; information geometry is physics. This is the final clarity of the Theory of Entropicity.

Summary and the Geomaga Trinity

In the Theory of Entropicity (ToE), the three fundamental structures of information geometry play roles analogous to the geometric structures of general relativity:

- the Fisher–Rao metric measures distinguishability of probability distributions and becomes the amplitude-geometry that generates emergent spacetime;
- the Fubini–Study metric measures distinguishability of quantum states and becomes the dynamical-geometry responsible for matter;
- the Amari–Cencov alpha-connections encode the dualistic affine structure of information flows and become the phase-geometry underlying gauge fields.

Together, these three structures form the entropic-geometric triad from which spacetime, matter, and electromagnetism emerge.

The relationship forged by the Theory of Entropicity (ToE) is as follows:

The Geomaga Trinity (TGT) of the Theory of Entropicity

Fisher–Rao

↓

becomes spacetime metric after emergence

Fubini–Study

↓

becomes matter sector after emergence

α -connections

↓

become gauge connections after emergence

The geo-ma-ga (GMG) trinity of Theory of Entropicity (ToE):

The Geomaga Trinity (TGT) of Theory of Entropicity (ToE).

This is the geometry-matter-gauge trinity of ToE: The geo-ma-ga (GMG) trinity of Theory of Entropicity (ToE): The Geomaga Trinity (TGT) of Theory of Entropicity (ToE).

Scholium: The Assignments Are Forced

Scholium §1. The Assignments Are Not Arbitrary

1. The assignments above in ToE are not arbitrary — they are forced by the type of geometry each structure encodes.

This is a claim of extraordinary strength, and it deserves correspondingly careful justification. In mathematics, when we say that a construction is “forced,” we mean that it is the unique construction satisfying certain natural conditions — there are no alternatives. The assignments of information-geometric structures to physical sectors in the Theory of Entropicity are forced in this precise sense: given the mathematical properties of each information-geometric structure and the physical requirements of each physical sector, there is exactly one consistent assignment.

The argument proceeds by examining each structure in turn, listing its mathematical properties, showing that these properties match the physical requirements of one and only one physical sector, and demonstrating that they are incompatible with the requirements of the other two sectors. The result is a unique, forced assignment — a tripartite correspondence that admits no permutations and no alternatives.

In the language of category theory, this forced assignment can be understood as a *natural transformation* between the category of information-geometric structures (with morphisms given by structure-preserving maps) and the category of physical structures (with morphisms given by physical symmetries). A natural transformation is one that is defined by the abstract structural properties of the objects, not by arbitrary choices. The Geomaga Trinity is natural in this sense: it is defined by what each information-geometric structure *is*, not by a choice of how to use it.

Scholium §2. Why Fisher-Rao Must Become the Spacetime Metric

2. Why Fisher-Rao must become the spacetime metric.

The Fisher-Rao metric has the following five defining properties:

Properties of the Fisher-Rao Metric

1. **Monotonicity** under coarse-graining (Markov morphisms)
2. **Invariance** under reparametrization (diffeomorphism-invariance)
3. **Uniqueness** (Cencov's theorem: unique up to a constant)
4. **Positivity** (positive-definite Riemannian metric)

5. **Distinguishability** (measures statistical distance between distributions)

Each of these properties has a direct physical counterpart in the requirements for a spacetime metric:

Matching Physical Requirements

1. **Coarse-graining** → Physical spacetime distances cannot increase under loss of microscopic information (the second law of thermodynamics in the gravitational sector). When microscopic degrees of freedom are integrated out, the emergent macroscopic distances can only decrease or remain the same, never increase. This is the physical analogue of Fisher-Rao monotonicity.

2. **Reparametrization invariance** → General covariance (the fundamental symmetry of general relativity). The spacetime metric must be independent of the choice of coordinates, just as the Fisher-Rao metric is independent of the parametrization of the statistical manifold.

3. **Uniqueness** → The spacetime metric is the unique gravitational degree of freedom (Lovelock's theorem ensures the uniqueness of the Einstein field equations). The uniqueness of the Fisher-Rao metric guarantees that the emergence map produces a unique spacetime metric, not a family of alternatives.

4. **Positivity** → Physical distances are positive (or, more precisely, the spatial part of the spacetime metric is positive-definite). The positivity of the Fisher-Rao metric ensures that the emergent spatial distances are

positive.

5. Distinguishability → Physical distances measure distinguishability of spacetime events. Two events at the same spacetime location are indistinguishable; two events at different locations are distinguishable by the spacetime interval between them. This is exactly what the Fisher-Rao metric measures on the statistical manifold.

This is why Fisher-Rao is the only information metric that can serve as the pre-geometric origin of spacetime. No other information metric has these properties. Therefore: Fisher-Rao → spacetime metric is not a choice in ToE. It is a necessity in the formulation of ToE.

To elaborate on why the match is unique: consider the alternative possibility that the Fubini-Study metric might serve as the spacetime metric. The Fubini-Study metric is defined on complex projective space CP^n , not on a manifold of probability distributions. It does not satisfy Cencov's uniqueness theorem (which applies only to statistical manifolds). It has constant positive holomorphic sectional curvature, which would force the emergent spacetime to be a space of constant positive curvature — ruling out flat spacetime, Schwarzschild black holes, and the entire phenomenology of general relativity. The Fubini-Study metric is therefore mathematically incompatible with the role of spacetime metric.

Similarly, the α -connections cannot serve as the spacetime metric because they are not metrics at all — they are affine connections. An affine connection defines parallel transport and curvature, but it does not define distances. A spacetime metric must define distances (the spacetime interval); an affine

connection cannot. The α -connections are thus the wrong *type* of geometric object to serve as the spacetime metric.

Scholium §3. Why Fubini-Study Must Become the Matter Sector

3. Why Fubini-Study must become the matter sector.

The Fubini-Study metric has the following defining properties:

Properties of the Fubini-Study Metric

1. **Dynamical** (measures change in quantum/entropic states)
2. **Kähler** (compatible complex, symplectic, and Riemannian structure)
3. **Projective** (defined on $\mathbb{CP}^n$, the space of rays, not vectors)
4. **Bounded curvature** (constant holomorphic sectional curvature)
5. **Quantum distinguishability** (measures distance between quantum states)

Each of these has a direct match to the matter sector:

Matching Physical Requirements

1. **Dynamical** → Matter is inherently dynamical — it propagates, oscillates, and evolves. The Fubini-Study metric measures the rate of state change, which in the emergent spacetime becomes the kinetic energy of matter fields.
2. **Kähler** → The matter sector requires compatibility of complex, symplectic, and metric structures — this is precisely the structure of

quantum field theory, where fields are complex-valued and the symplectic structure defines the Poisson bracket (and hence the quantum commutator).

3. **Projective** → Matter fields in quantum mechanics are rays, not vectors (overall phase is unobservable). The projective nature of the Fubini-Study metric respects this fundamental feature.

4. **Bounded curvature** → The energy density of matter is bounded (by the Planck density, in the quantum gravity regime). The bounded curvature of the Fubini-Study metric provides a natural mechanism for this bound.

5. **Quantum distinguishability** → Matter states are distinguished by quantum measurements. The Fubini-Study metric encodes this distinguishability.

Therefore: Fubini-Study → matter sector is forced by its meaning.

Could the Fisher-Rao metric serve as the matter sector instead? No. The Fisher-Rao metric is a metric on the space of probability distributions, not on the space of quantum states. It does not have Kähler structure (it is real, not complex). It does not have the projective property (probability distributions are normalized vectors in L^1 , not rays in a Hilbert space). It measures classical statistical distinguishability, not quantum state change. The Fisher-Rao metric is the wrong type of object for the matter sector, just as it is the right type for the spacetime sector.

Could the α -connections serve as the matter sector? No. As noted above, the α -connections are affine connections, not metrics. The matter sector requires a metric (to define kinetic energy, which is a quadratic form on velocities). An

affine connection defines parallel transport, not inner products. The α -connections cannot define the energy density or the stress-energy tensor that are characteristic of the matter sector.

Scholium §5. Why α -Connections Cannot Be the Levi-Civita Connection

5. Why alpha-connections cannot be the Levi-Civita connection.

The distinction between α -connections and the Levi-Civita connection is sharp and fundamental:

Levi-Civita Connection Properties

- Torsion-free
- Metric-compatible ($\nabla g = 0$)
- Uniquely determined by the metric
- Defines parallel transport preserving inner products
- Curvature = Riemann curvature tensor (gravitational tidal forces)

α -Connection Properties (for $\alpha \neq 0$)

- Torsion-free
- NOT metric-compatible ($\nabla^{(\alpha)} g \neq 0$)
- NOT uniquely determined by the metric (depends on α and T_{abc})
- Parallel transport does NOT preserve inner products
- Curvature = statistical curvature (not gravitational tidal forces)

Therefore: α -connections cannot be the Levi-Civita connection and cannot produce Einstein's field equations. They are the wrong type of geometric object. They are affine connections on a statistical manifold, not metric connections on a Riemannian manifold.

This distinction is not merely technical; it is the fundamental mathematical reason why the α -connections must correspond to gauge fields rather than to gravity. In general relativity, gravity is encoded in the Levi-Civita connection of the spacetime metric — the unique metric-compatible, torsion-free connection. This connection preserves lengths and angles under parallel transport, which is the geometric expression of the equivalence principle (the laws of physics are the same in all freely falling frames). The α -connections do not preserve lengths under parallel transport; they allow the “size” of a statistical distribution to change as it is transported. This non-metric-compatible behavior is precisely the behavior of a gauge connection, which changes the phase of a charged field under parallel transport while preserving its modulus.

To produce Einstein's field equations, one needs the Riemann curvature tensor formed from the Levi-Civita connection of a metric. The curvature tensor of an α -connection is, in general, different from the Riemann curvature tensor: it contains additional terms involving the skewness tensor T_{abc} and its derivatives. These additional terms do not have the correct algebraic symmetries (the Bianchi identity, the pair symmetry) to enter into the Einstein field equations. They have the algebraic structure of a gauge field strength tensor, which is antisymmetric in two indices (like the electromagnetic field strength $F_{\mu\nu}$) rather than having the pair symmetries of the Riemann tensor. This algebraic mismatch is the final, conclusive proof that the α -connections cannot play the role of the gravitational connection and must instead play the role of gauge connections.

Scholium §6. The Deep Reason the Assignment Is Forced

6. The deep reason the assignment is forced.

Information Geometry Structure	What It Measures	Physical Meaning in ToE
Fisher-Rao metric	amplitude distinguishability	spacetime geometry
Fubini-Study metric	dynamical distinguishability	matter/energy
α -connections	phase/affine duality	gauge fields

This is the geometry-matter-gauge trinity of ToE. It is not arbitrary. It is not aesthetic. It is not chosen. It is forced by the mathematical nature of each structure.

The depth of this forced assignment can be appreciated by considering it from the perspective of mathematical naturality. In category theory, a natural transformation between two functors is one that commutes with all morphisms in the underlying categories. Applied to the present context: the functor from the category of information-geometric structures to the category of physical structures must commute with all natural maps (Markov morphisms on the statistical side, diffeomorphisms on the physical side). The Fisher-Rao metric is the unique object in the information-geometric category that is invariant under Markov morphisms (Cencov’s theorem), just as the spacetime metric is the unique gravitational object in the physical category that is invariant under diffeomorphisms (general covariance). The natural transformation must therefore map Fisher-Rao to spacetime metric. Similarly, the Fubini-Study metric is the unique Kähler metric on the quantum state space invariant under unitary transformations, just as the matter sector is the unique physical sector that respects the unitary

symmetry of quantum mechanics. The natural transformation maps Fubini-Study to matter. And the α -connections are the unique dual affine structures on the statistical manifold, just as gauge connections are the unique additional geometric structures (beyond the metric) in physical spacetime. The natural transformation maps α -connections to gauge fields. The assignment is not merely “forced” in the informal sense; it is *naturally* forced, in the precise category-theoretic sense.

Scholium §7. The Final Clarity of ToE

7. The final clarity of ToE.

The Stable Assignment

alpha-connections are affine, not metric $\rightarrow$ gauge

Fisher-Rao is metric, not affine $\rightarrow$ spacetime

Fubini-Study is dynamical, not metric $\rightarrow$ matter

This is the correct, final, stable assignment of information geometry in ToE.

The word “stable” in this declaration deserves emphasis. In mathematics, a result is stable if it does not change under small perturbations of the hypotheses. The Geomaga Trinity is stable in this sense: small modifications to the mathematical properties of the three information-geometric structures, or small variations in the physical requirements of the three physical sectors, do not change the assignment. The Fisher-Rao metric is robustly the right object for spacetime (any small deformation of the Fisher-Rao metric that preserved its essential properties would still be mapped to the spacetime metric, since Cencov’s theorem is itself a robustness result). The Fubini-

Study metric is robustly the right object for matter. The α -connections are robustly the right objects for gauge fields. The assignment is stable, unique, and final.

Scholium §8. ToE Summary

8. ToE Summary.

“In the Theory of Entropicity, the Fisher-Rao metric provides the amplitude-geometry that becomes spacetime, the Fubini-Study metric provides the dynamical-geometry that becomes matter, and the Amari-Cencov alpha-connections provide the phase-geometry that becomes gauge fields. This assignment is not arbitrary; it is uniquely determined by the mathematical meaning of each structure.”

This is the truth of Obidi’s formulation of the Theory of Entropicity (ToE).

Additional Scholium Remarks: Uniqueness of the Geomaga Trinity

The uniqueness of the Geomaga Trinity invites comparison with other uniqueness results in theoretical physics. The most celebrated such result is *Lovelock’s theorem* (1971), which states that in four spacetime dimensions, the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - (1/2)Rg_{\mu\nu}$ is the unique symmetric, divergence-free, second-rank tensor that is a function of the metric and its first and second derivatives only. This theorem guarantees that the Einstein field equations are the unique gravitational field equations of their type — there are no alternatives. Lovelock’s theorem is what gives general relativity its

inevitability: given the symmetry requirements and the derivative order, the field equations are forced.

The Geomaga Trinity enjoys an analogous inevitability. Given the mathematical properties of the three information-geometric structures and the physical requirements of the three physical sectors, the assignment is forced. There are no alternatives. Just as Lovelock's theorem eliminates the possibility of alternative gravitational field equations (within its assumptions), the mathematical structure of information geometry eliminates the possibility of alternative assignments of information-geometric structures to physical sectors (within the framework of the Theory of Entropicity).

Consider the six possible permutations of the three-to-three assignment:

5. **Fisher-Rao $\rightarrow$ spacetime, Fubini-Study $\rightarrow$ matter, α -connections $\rightarrow$ gauge** (the Geomaga Trinity)
6. Fisher-Rao $\rightarrow$ spacetime, Fubini-Study $\rightarrow$ gauge, α -connections $\rightarrow$ matter: FAILS because α -connections are not a metric and cannot define the kinetic energy of matter, and Fubini-Study has Kähler structure incompatible with gauge connections.
7. Fisher-Rao $\rightarrow$ matter, Fubini-Study $\rightarrow$ spacetime, α -connections $\rightarrow$ gauge: FAILS because Fubini-Study has constant curvature on CP^n , incompatible with the variable curvature of physical spacetime, and Fisher-Rao is not Kähler as required by the matter sector.
8. Fisher-Rao $\rightarrow$ matter, Fubini-Study $\rightarrow$ gauge, α -connections $\rightarrow$ spacetime: FAILS because α -connections are not metric-compatible, so they cannot define spacetime distances.
9. Fisher-Rao $\rightarrow$ gauge, Fubini-Study $\rightarrow$ spacetime, α -connections $\rightarrow$ matter: FAILS for the same reasons as permutations 3 and 4.

10. Fisher-Rao $\rightarrow$ gauge, Fubini-Study $\rightarrow$ matter, α -connections $\rightarrow$ spacetime: FAILS because Fisher-Rao is a metric (not an affine connection) and cannot serve as a gauge connection, and α -connections cannot serve as the spacetime metric.

Only permutation 1 — the Geomaga Trinity — is mathematically consistent. The assignment is unique. It is forced. It is the truth of the Theory of Entropicity.

Concluding Remarks

This Preamble has established, at monograph-grade depth, the conceptual and philosophical foundations for the physical status of information-geometric structures in the Theory of Entropicity. The argument is structured as a series of eight interlocking claims, each building upon the preceding ones:

11. Einstein established the precedent that mathematical geometry can be identified with physical reality (Section 1).
12. The Fisher-Rao, Fubini-Study, and α -connection structures measure precisely the quantities that physics requires: distinguishability, dynamical change, and information flow (Section 2).
13. The Theory of Entropicity makes the same kind of ontological leap as Einstein, but at a deeper level: information geometry is not a description of physics but is the substrate of physics (Section 3).
14. The Fisher-Rao metric becomes the spacetime metric via the emergence map, with the conversion factor $\lambda \approx 10^{-47}$ fixed by Cencov's theorem and the entropic curvature gap (Section 4).

15. The Fubini–Study metric becomes the matter sector, with mass as resistance to entropic change and energy as Fubini–Study curvature (Section 5).
16. The α -connections become electromagnetism, with the phase geometry of the entropic field generating the U(1) gauge structure (Section 6).
17. The three-sector unification (spacetime, matter, gauge) from a single entropic field is deeper than any previous unification in the history of physics (Section 7).
18. The information-geometric structures do not need to be “forced” to be physical; they naturally measure what physics requires (Section 8).

The Scholium then demonstrated that the assignment of information-geometric structures to physical sectors is not arbitrary but is uniquely forced by the mathematical properties of each structure. Only the Geomaga Trinity — Fisher–Rao to spacetime, Fubini–Study to matter, α -connections to gauge fields — is mathematically consistent. All five alternative permutations fail.

The reader is now prepared to enter the body of Letter IE, where the mathematical implementation of these ideas is carried out in detail: the Obidi Action is constructed, the field equations are derived, the Curvature Transfer Theorem is proved, and the recovery of general relativity, quantum mechanics, and Maxwell’s electrodynamics from the single framework of information geometry is demonstrated. The present Preamble has provided the *why*; the body of Letter IE provides the *how*.

— End of Preamble —